

ADITHYA A RAO

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Education

PhD in Physics

Indian Institute of Science Education and Research (IISER), Pune

2025 –

- Theoretical High Energy Physics.
- Advisor – Prof. Sachin Jain.
- Courses taken – Quantum Field Theory I & II, General Relativity, Advanced Gravitation, Group Theory Methods in Physics, Advanced Statistical Mechanics
- CGPA – 10 /10

5-year Integrated MSc | *Physics*

National Institute of Technology (SVNIT), Surat, India

2019 – 2024

- Master’s Thesis – *Gribov Problem and Stochastic Quantization*
 - Advisors – Prof. Laurent Baulieu & Dr. Vikash K. Ojha
 - CGPA – 9.77 /10
 - **Institute Gold Medalist**
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Publications

A. Bala, S. Jain, D. K. S, and **A. A. Rao**, “Super-Grassmannians for $\mathcal{N} = 2$ to 4 SCFT₃: From AdS₄ Correlators to $\mathcal{N} = 4$ SYM scattering Amplitudes,” arXiv (Cornell University), Apr. 2026, doi: [10.48550/arxiv.2604.07503](https://doi.org/10.48550/arxiv.2604.07503).

A. Bala, S. Jain, D. K. S, and **A. A. Rao**, “The $\mathcal{N} = 1$ Super-Grassmannian for CFT₃ and a Foray on AdS and Cosmological Correlators,” arXiv (Cornell University), Apr. 2026, doi: [10.48550/arxiv.2604.07446](https://doi.org/10.48550/arxiv.2604.07446).

V. K. Ojha, **A. A. Rao**, and S. D. Pathak, “Estimating the Age of Universe via Scalar Field,” in Proceedings of the XXV DAE-BRNS High Energy Physics (HEP) Symposium 2022, 12–16 December, Mohali, India, vol. 304, Singapore: Springer Nature Singapore, 2024, pp. 1015–1016. doi:10.1007/978-981-97-0289-3.272. For direct access: [Link](#)

Conference Presentations

V. K. Ojha, **A. A. Rao**, and S. D. Pathak. (2022), “*Estimating the Age of Universe via Interacting Tachyonic Scalar Field*”. XXV DAE-BRNS High Energy Physics Symposium 2022. [Poster presentation](#).

Honors and Awards

- 2025** All India Rank 116, CSIR-NET Physical Sciences Paper (JRF Qualified).
- 2025** All India Rank 64 (99.7 percentile), GATE — Physics paper.
- 2025** Institute Gold Medal, National Institute of Technology, Surat, India.
- 2024** SVNIT Alumni Association Gold Medal for securing first rank in MSc Physics.
- 2024** Young Scientist Participant, 73rd Lindau Nobel Laureate Meeting in Physics, selected as a part of the Indian Delegation, one of 16 Indian and 640 international young scientists.
- 2023** DAAD-WISE award, fully funded research project at Friedrich Schiller University Jena, Germany, one of 130 Indian students.
- 2021** IASc, INSA, NASI – Summer Research Fellowship Program award, fully funded research project at PRL Ahmedabad, India, one of 100 Indian students.
- 2019** Department of Science and Technology, Government of India – INSPIRE (INnovation in Science Pursuit for Inspired REsearch) award. Awarded to the top 1% students of the country.

Relevant Technical Skills

Programming Languages: C++, C, Python (PyTorch, NumPy, TensorFlow), MATLAB, Julia, FORTRAN

Data Analysis Tools: ROOT

Computer Algebra Systems: Mathematica, SymPy

Typesetting Tools: L^AT_EX

Version Control: Git, GitHub

Additional: Linux, Bash scripting, High Performance Computing (Basic)

Outreach Activities

- 2022–23** Head, [Physics Club of NIT Surat](#) (Member and web-developer 2020–24), a club actively conducting events to impart knowledge and inculcate interest in physics in the young minds and the general public.
- 2020–** Author, [thehavok.com](#), writing science articles to make the scientific jargon accessible to even a non-academic.